

Task-Oriented Learning for Automatic EEG Denoising

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Abstract—Electroencephalography (EEG) denoising methods typically depend on manual intervention or clean reference signals. This work introduces a task-oriented learning framework for automatic EEG denoising with only task labels without clean reference signals. EEG recordings are first decomposed into components based on blind source separation (BSS) techniques. A learning-based selector then assigns a retention probability to each component, and the denoised signal is reconstructed as a probability-weighted combination. A downstream proxy-task model evaluates the reconstructed signal, with its task loss supervising the selector in a collaborative optimization scheme that relies solely on task labels, eliminating the need for clean EEG references. Experiments on three datasets, covering two paradigms and multiple noise conditions, demonstrate average gains of 2.59% in task accuracy and 0.80 dB in signal-to-noise ratio under standard signal-quality metrics. Further analyses demonstrate that the task-oriented learning framework is algorithm-agnostic, as it accommodates diverse decomposition techniques and network backbones for both the selector and the proxy model. These results indicate the proposed framework can denoise EEG using only task labels, supporting its potential utility in neuroscience research and EEG-based interaction systems.

Index Terms—Electroencephalography analysis, signal denoise

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I. INTRODUCTION

Decoding and analyzing brain activity using neuroimaging has long been investigated for its potential to rehabilitate disabilities and improve healthy function [1]–[4]. Within available modalities, the electroencephalogram (EEG) is particularly attractive because it offers a favorable balance among cost, safety, and temporal resolution [5]. However, EEG signals are intrinsically weak and highly susceptible to contamination. Noise in EEG recordings originates from three primary sources: physiological artifacts, instrumental factors, and environmental interferences [6], [7]. Such noise reduces the signal-to-noise ratio (SNR) and can obscure task-related information.

These observations underscore the need for advanced denoising methods to improve EEG quality and increase decoding reliability [8]. Despite extensive efforts to acquire high-quality EEG data, the inherent coupling of nonlinear noise with subtle neural components complicates the recovery of clean signals [6]. Research spanning nearly five decades has yet to reach consensus on an optimal denoising algorithm for practical deployment [6]. Moreover, a recent overview of EEG applications emphasized artifact detection and removal as a top priority in EEG signal processing [8].

Current EEG denoising techniques can generally be classified into traditional signal-processing-based and deep-learning-based methods. Traditional methods leverage signal-processing techniques such as filtering [9], regression [10], and BSS (e.g., independent component analysis (ICA) [11], principal component analysis (PCA) [12], canonical correlation analysis (CCA) [13]). However, a key limitation of these traditional methods is the reliance on expert intervention. Filtering and regression approaches typically depend on handcrafted features derived from statistical analysis, whereas BSS methods require experts to inspect and select decomposed noisy components manually. Such dependence on expert knowledge increases complexity and limits practicality, particularly for online or real-time applications.

Deep-learning-based approaches implement an end-to-end pipeline that requires neither pre-defined indices nor manual intervention. These methods can be categorized into two main types: autoencoder-based [14]–[18] and generative adversarial network (GAN)-based models [19]–[21]. Autoencoder-based models learn a mapping from noisy to clean EEG signals via various architectures, including convolutional neural networks (CNN) [14], [15], transformers [16], [17], and recurrent neural networks (RNN) [18]. GAN-based methods comprise a generator G and a discriminator D [19]–[21]; G produces candidate

clean EEG signals, while D evaluates whether an input signal is real or generated. Training proceeds adversarially to optimize G such that D cannot distinguish generated denoised signals from true clean EEG data.

However, deep-learning-based methods also exhibit several limitations. First, the training process often requires clean EEG signals: autoencoder-based methods typically need paired clean-noisy EEG signals to learn the transformation [14]–[18], while GAN-based methods require a set of clean EEG data to model the clean signals distribution [19]–[21]. Such clean EEG data are difficult to acquire in practice. In addition, inevitable distribution shifts between the training set and the test set may introduce artifacts or suppress task-related information during the denoising process. Thus, there remains room for improving deep-learning-based methods.

To address the limitations noted above, a task-oriented learning framework for automatic EEG denoising is proposed. The approach combines classical BSS signal processing with deep learning. Raw EEG is first decomposed into components based on BSS algorithms. Then, a learning-based selector identifies informative components and reconstructs a denoised signal as a probability-weighted combination. Selector training is guided by a proxy-task model in a collaborative optimization scheme, enabling selection of task-relevant components without requiring component-level labels. By combining the strengths of BSS-based and learning-based methods, the proposed framework enables automatic EEG denoising without clean reference signals and reduces the risk of introducing artifacts. Moreover, the task-oriented design explicitly enhances the salience of task-related information within the EEG signals, which is the primary motivation for EEG denoising.

The primary contributions of this study are as follows:

- A task-oriented learning framework for automatic EEG denoising is introduced, which is trained solely on task-related labels, explicitly retaining task-relevant information without relying on clean reference signals.
- A collaborative optimization scheme is proposed that integrates signal-processing and deep-learning techniques, where a proxy task guides the selector to recombine decomposed components. This design enables automatic denoising while avoiding introducing additional artifacts.
- Experimental evaluation on three datasets and two paradigms demonstrates improvements in signal quality (SNR: 0.80 dB $\uparrow$) and task-relevant metrics (accuracy: 2.59% $\uparrow$); compatibility with various BSS methods and learning-based backbones confirms the proposed task-oriented learning framework is algorithm-agnostic.

The remainder of this paper is organized as follows. Section II reviews related work. Section III details the proposed approach. Section IV presents the experimental results and ablation studies. Section V discusses the proposed framework in terms of signal properties and task-relevant information preservation. Section VI concludes the paper.

II. RELATED WORK

A. Traditional Signal-processing-based Methods

EEG denoising is a fundamental signal-processing problem. Numerous classical signal processing methods have been de-

veloped, among which BSS is a representative technique [11]–[13], [22]–[25]. BSS decomposes EEG signals into constituent components via matrix decomposition methods such as ICA [11], PCA [12], and CCA [13]. However, identifying which decomposed components carry task-related information and excluding noise remains challenging. Although automated component selection frameworks have been proposed [11], [25], standard pipelines and analyses still depend heavily on manual selection.

Another traditional signal-processing approach treats EEG signals as composite and applies techniques such as wavelet transform [26] and adaptive filtering [9] for denoising [9], [10], [26]. However, these methods typically rely on predefined, handcrafted rules or features. In summary, traditional signal-processing-based approaches formulate the denoising task in an interpretable manner and, when properly configured, do not introduce spurious artifacts. Nevertheless, they still require expert intervention and handcrafted rules, which limit their practical applications.

B. Deep-learning-based Methods

Deep-learning-based approaches offer an automated way for EEG denoising. These approaches are categorized into autoencoder-based and GAN-based methods. Autoencoder-based models are trained to map noisy EEG inputs to denoised outputs, via various architectures to capture spatial-temporal representations such as CNN [14], [15], [27]–[29], RNN [18], [30], and transformer [16], [17], [31]–[33]. GAN-based methods do not perform direct regression from noisy to clean signals [19]–[21]. Instead, a generator produces candidate clean signals, while a discriminator distinguishes between real and generated signals. The adversarial training objective encourages the generator to approximate the distribution of clean EEG signals rather than memorize individual examples, yielding outputs that are similar to the true clean signals.

Although deep-learning-based methods can denoise EEG signals automatically, their training process often relies on clean EEG data, which poses a significant challenge because high-quality signals are difficult to obtain. Moreover, since denoising is data-driven, distribution shifts between training and test data may introduce new artifacts or suppress task-related information.

III. METHOD

A. Task-Oriented Learning Framework

The proposed task-oriented learning framework for automatic EEG denoising is model-agnostic (Fig.1) and can be realized with different BSS decomposition algorithms & network architectures. Raw EEG data $\mathbf{X}$ is first decomposed into N components $\{\mathbf{C}_i\}_{i=1}^N$ using a BSS algorithm:

$$\mathbf{X} = g(\mathbf{X}) = \sum_{i=1}^N \mathbf{C}_i \quad (1)$$

where $g(\cdot)$ denotes the decomposition operator. Each component $\mathbf{C}_i$ is then assigned a selection score p_i by a selector function $f_s(\cdot)$:

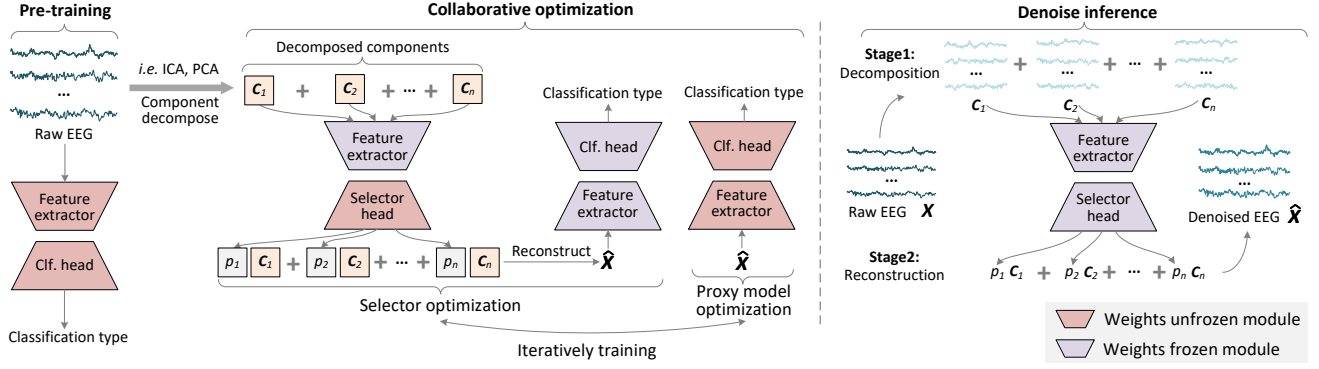


Fig. 1. Overview of the proposed task-oriented learning framework for automatic EEG denoising. (i) Pre-training: the feature extractor and classification (Clf.) head are trained on a supervised proxy classification task using raw EEG. (ii) Collaborative Optimization: EEG is decomposed into components by BSS methods (e.g., ICA/PCA); the selector predicts component-retention probabilities and is optimized collaboratively with the proxy-task model. (iii) Denoise inference: the trained selector alone weights task-relevant components to reconstruct the denoised EEG.

$$p_i = f_s(C_i) \quad (2)$$

where $p_i \in [0, 1]$ represents the probability that C_i is informative rather than noisy. The denoised EEG signal $\hat{X}$ is reconstructed as a weighted sum of components:

$$\hat{X} = \sum_{i=1}^N p_i C_i \quad (3)$$

since component-level labels are unavailable, a proxy downstream task is introduced to guide selector training. Let y denote the ground-truth label and $f_p(\cdot)$ the proxy-task model. The selector loss is given by:

$$\mathcal{L} = c(f_p(\hat{X}), y) \quad (4)$$

where $c(\cdot, \cdot)$ denotes the designed loss function. Minimizing $\mathcal{L}$ encourages retention of components that enhance proxy-task performance and suppression of task-irrelevant noise.

B. EEG Decomposition

Although various decomposition methods exist for EEG analysis, the result can generally be expressed as:

$$X = g(X) = AB^T + \epsilon \quad (5)$$

where $X \in \mathbb{R}^{C \times T}$ denotes the raw EEG signal with C electrodes and T time points, $A \in \mathbb{R}^{C \times M}$ and $B \in \mathbb{R}^{T \times M}$ are factor matrices with $M \leq C$, and ϵ is the residual matrix. Let $a_i \in \mathbb{R}^{C \times 1}$ and $b_i \in \mathbb{R}^{T \times 1}$ denote the i th columns of A and B . Then Eq. 5 admits the component-wise form:

$$X = \sum_{i=1}^M a_i b_i^T + \epsilon = \sum_{i=1}^M C_i + C_{M+1} = \sum_{i=1}^N C_i \quad (6)$$

where

$$C_i = a_i b_i^T \quad (i = 1, \dots, M), \quad C_{M+1} = \epsilon, \quad N = M + 1.$$

each component C_i is the outer product of a spatial pattern a_i and a temporal pattern b_i , so that each row of C_i encodes the signal distribution across electrodes for that component.

Algorithm 1 : Training procedure

Require: Dataset $\mathcal{D} = \{(X_j, y_j)\}$, epochs E , batch size S
Ensure: f_s predicts the retention probability of each component

- 1: — **Pre-training stage** —
- 2: Initialize the proxy model f_p (Eq. 8)
- 3: Pre-train f_p on $\mathcal{D}$ using supervised learning
- 4: Initialize selector head h_s and shared q to form f_s (Eq. 7)
- 5: — **Collaborative optimization stage** —
- 6: **for** epoch = 1 to E **do**
- 7: **repeat**
- 8: Sample a batch $B = \{(X_k, y_k)\}_{k=1}^S$ from $\mathcal{D}$
- 9: Decompose each $X \in B$ via $g(\cdot)$ (Eq. 5 and 6)
- 10: Reconstruct EEG signals (Eq. 2 and 3)
- 11: — **Update Selector** —
- 12: Unfreeze h_s ; freeze q and h_p
- 13: Optimize h_s with AdamW using the loss in Eq. 4
- 14: — **Update Proxy Model** —
- 15: Unfreeze q and h_p ; freeze h_s
- 16: Optimize f_p with AdamW using the loss in Eq. 4
- 17: **until** All samples in $\mathcal{D}$ have been sampled.
- 18: **end for**

Hence, the operator $g(\cdot)$ in Eq. 1 can be treated as two steps: the initial matrix factorization in Eq. 5 and the post-processing into accumulated spatial-temporal components in Eq. 6.

C. Collaborative Optimization

1) *Optimization Process:* The optimization process involves two models: a selector that identifies task-relevant components and a proxy-task model that provides task supervision. Since component labels are unavailable, guidance for the selector is provided by performance on the proxy task. This design is motivated by the assumption that task-relevant components contribute useful information, whereas task-irrelevant components introduce noise. Therefore, the performance of the reconstructed EEG signals on the proxy task is used as the evaluation metric for the selector model.

The selector f_s comprises two modules:

$$f_s = h_s \circ q \quad (7)$$

where q denotes the feature extractor shared with the proxy-task model, and h_s represents the selector head, which outputs the probability that an input component is not noisy. Although the selector and proxy downstream classification tasks differ, both depend on the same extracted task-related representations. Thus, the feature extractor is shared by the two models. The proxy-task model f_p is defined as:

$$f_p = h_p \circ q \quad (8)$$

where h_p denotes the classification head for the downstream task. Both h_s and h_p are implemented as fully connected layers that map the extracted features from the EEG or the decomposed components to output probabilities.

The optimization procedure is presented in Algorithm 1, which consists of two stages: pre-training and collaborative optimization. During pre-training, the feature extractor q is initialized via supervised learning to establish a robust shared representation for both the selector and proxy-task model. This process lets the selector focus on the discriminative features related to the task.

The second stage performs collaborative optimization between the selector and the proxy-task model. The selector is updated under proxy-task supervision to favor components that improve downstream task accuracy. The proxy-task model is then refined by minimizing the classification loss computed on the reconstructed EEG signals. This collaborative scheme guides the selector to retain task-relevant components while suppressing noise, achieving EEG denoising.

2) *Convergence Analysis*: The following hypotheses are adopted to analyze the convergence of the optimization:

- **H1**: $\inf \mathcal{L} > -\infty$.
- **H2**: The loss $c(\cdot, \cdot)$ and h_p , q , and h_s are continuously differentiable with Lipschitz-continuous gradients on a bounded domain; $\hat{X}$ is continuously differentiable with respect to the model parameters.
- **H3**: Gradients are bounded and the variance of the mini-batch stochastic noise is bounded; features lie in a bounded set.
- **H4**: Each module update does not increase the loss, either in full-batch form or in expectation in stochastic settings.

In the implementation, these hypotheses are encouraged as follows. **H1** holds because $\mathcal{L}$ is a cross-entropy loss and is therefore bounded below by zero. To promote **H2**, ELU activations are used, max-norm constraints are imposed on layer weights, and gradient clipping is applied. For **H3**, gradient clipping yields bounded gradients, while the feature extractor operating with bounded weights on finite-energy inputs ensures bounded feature vectors. Regarding **H4**, AdamW with the AMSGrad variant and a cosine learning-rate decay is employed to stabilize updates so that the loss is non-increasing in expectation.

Based on the hypotheses, a discussion about the convergence is carried out. Define the collaborative objective as:

$$\mathcal{L}(\theta_s, \theta_p) \triangleq c(f_p(\hat{X}(\theta_s, \theta_p)), y) \quad (9)$$

Algorithm 2 : Inference procedure for EEG denoising

Require: Raw EEG signal X

Ensure: Denoised EEG signal $\hat{X}$

1: — **Step 1: Decompose the EEG signal** —

2: Compute components $\{C_i\}_{i=1}^N \leftarrow g(X)$ (Eq. 1 and 6)

3: — **Step 2: Infer component retention probabilities** —

4: **for** $i = 1$ to N **do**

5: $p_i \leftarrow f_s(C_i)$ (Eq. 2)

6: **end for**

7: — **Step 3: Reconstruct the EEG signal** —

8: $\hat{X} \leftarrow \sum_{i=1}^N p_i C_i$ (Eq. 3)

where θ_s denotes the parameters of h_s , and θ_p collects the parameters of q and h_p . Under the hypotheses, $\mathcal{L}$ is block-wise continuously differentiable, and for fixed θ_p (or θ_s), $\nabla_{\theta_s} \mathcal{L}$ (or $\nabla_{\theta_p} \mathcal{L}$) is Lipschitz continuous on a bounded domain.

The update can be considered iteratively, step A (selector update), for some α_s :

$$\theta_s^{t+1} = \theta_s^t - \alpha_s \nabla_{\theta_s} \mathcal{L}(\theta_s^t, \theta_p^t), \quad \alpha_s \in (0, 2/\beta_s) \quad (10)$$

and Step B (proxy-model update), for some α_p :

$$\theta_p^{t+1} = \theta_p^t - \alpha_p \nabla_{\theta_p} \mathcal{L}(\theta_s^{t+1}, \theta_p^t), \quad \alpha_p \in (0, 2/\beta_p) \quad (11)$$

where α_s and α_p are the learning rates for each step, $\beta_s, \beta_p > 0$, $\mathcal{L}$ is β_s -smooth in θ_s with θ_p fixed and β_p -smooth in θ_p with θ_s fixed (block-wise Lipschitz gradients), and t is the iteration index. In Step A, θ_p is treated as constant, while in Step B, θ_s is constant.

By the descent lemma for a β_s -smooth function and Step A,

$$\begin{aligned} \mathcal{L}(\theta_s^{t+1}, \theta_p^t) &\leq \mathcal{L}(\theta_s^t, \theta_p^t) + \langle \nabla_{\theta_s} \mathcal{L}(\theta_s^t, \theta_p^t), \theta_s^{t+1} - \theta_s^t \rangle \\ &\quad + \frac{\beta_s}{2} \|\theta_s^{t+1} - \theta_s^t\|^2 \\ &= \mathcal{L}(\theta_s^t, \theta_p^t) - \left(\alpha_s - \frac{\beta_s}{2} \alpha_s^2 \right) \|\nabla_{\theta_s} \mathcal{L}(\theta_s^t, \theta_p^t)\|^2. \end{aligned} \quad (12)$$

Because $\alpha_s \in (0, 2/\beta_s)$ or satisfies an Armijo condition, the coefficient $\mu_s \triangleq \alpha_s - \frac{\beta_s}{2} \alpha_s^2$ is strictly positive, yielding:

$$\mathcal{L}(\theta_s^{t+1}, \theta_p^t) \leq \mathcal{L}(\theta_s^t, \theta_p^t) - \mu_s \|\nabla_{\theta_s} \mathcal{L}(\theta_s^t, \theta_p^t)\|^2 \quad (13)$$

similarly, holding θ_s^{t+1} fixed in Step B and using β_p -smoothness in θ_p

$$\mathcal{L}(\theta_s^{t+1}, \theta_p^{t+1}) \leq \mathcal{L}(\theta_s^{t+1}, \theta_p^t) - \mu_p \|\nabla_{\theta_p} \mathcal{L}(\theta_s^{t+1}, \theta_p^t)\|^2 \quad (14)$$

where $\mu_p \triangleq \alpha_p - \frac{\beta_p}{2} \alpha_p^2 > 0$. Combine Eqs. (13) and (14):

$$\begin{aligned} \mathcal{L}(\theta_s^{t+1}, \theta_p^{t+1}) &\leq \mathcal{L}(\theta_s^t, \theta_p^t) - \mu_s \|\nabla_{\theta_s} \mathcal{L}(\theta_s^t, \theta_p^t)\|^2 \\ &\quad - \mu_p \|\nabla_{\theta_p} \mathcal{L}(\theta_s^{t+1}, \theta_p^t)\|^2. \end{aligned} \quad (15)$$

Thus $\mathcal{L}(\theta_s^{t+1}, \theta_p^{t+1}) \leq \mathcal{L}(\theta_s^t, \theta_p^t)$ holds for all t . Moreover, by **H1**, $\mathcal{L}$ has a finite lower bound; hence, the monotone bounded sequence $\{\mathcal{L}(\theta_s^t, \theta_p^t)\}$ converges.

D. EEG Denoising Inference

Following training, raw EEG signals are denoised in two stages (Fig. 1, right). Stage 1 (Decomposition): the operator $g(\cdot)$ decomposes each raw EEG signal $\mathbf{X}$ into a set of components $\{\mathbf{C}_i\}$. Stage 2 (Reconstruction): the selector predicts the retention probability of each component, and the denoised EEG signal $\hat{\mathbf{X}}$ is reconstructed as a probability-weighted combination of the components for downstream applications. The detailed process can be found in Algorithm 2.

IV. EXPERIMENTS

The experiments are designed for the following questions:

- Does the framework work for different paradigms?
- Does the framework work for different noise?
- Does the framework achieve a competitive performance with the previous EEG denoising algorithms?
- Does the framework retain task-related information during the denoising process?

A. Dataset Description

Three EEG datasets from two paradigms are used in this study, including two publicly available datasets [34], [35] and one self-collected dataset [36].

- Steady-State Visual Evoked Potential (SSVEP) dataset [35]: a twelve-class SSVEP dataset containing 9 subjects, with 15 trials of 4.15 s per class, acquired by an 8-channel EEG system at 256 Hz.
- Motor imagery dataset [34]: a two-class motor imagery dataset containing 52 subjects, with 100 trials of 3 s per class for each subject, acquired by a 64-channel EEG system at 512 Hz and downsampled to 160 Hz.
- Motor execution dataset [36], [37]: a self-collected four-class motor execution dataset containing 10 subjects, with 100 trials of 4 s per class for each subject, acquired by a 32-channel EEG system at 128 Hz.

Because motor imagery and motor execution share the same underlying physiological mechanism: event-related desynchronization/synchronization (ERD/ERS) in the sensorimotor cortex [38]. These two paradigms are grouped, whereas SSVEP is treated separately.

B. Experimental Details

1) Noise Considered in this Study:

a) *Simulated Physiological Artifacts*: Following prior work on simulated physiological artifacts [14], electrooculogram (EOG) and electromyogram (EMG) signals are linearly combined with EEG signals to emulate irregular eye blinks and muscle activity. To synthesize varying noise levels, SNR values are drawn uniformly from $[-5, 5]$ dB, consistent with previous EEG noise studies [39]. Since eye blinks and muscle artifacts typically have limited impact on occipital recordings associated with the SSVEP paradigm [35], these simulated physiological artifacts are excluded from the SSVEP dataset.

b) *Inherent Instrumental and Environmental Noise*: Raw EEG signals acquired during measurement are inherently contaminated by instrumental and environmental noise, which is inseparable from the neural activities.

2) *Evaluation Metrics*: The denoising algorithms are evaluated from two perspectives:

- The extent to which the denoising process preserves task-related information in the EEG signal.
- The effectiveness of the denoising process in reducing noise, i.e., improving signal quality.

a) *Task-related Information Evaluation*: A key motivation for EEG denoising is to enhance the salience of task-related information. To assess this aspect directly, downstream-task performance using denoised signals is compared with that obtained from raw signals. Since the experimental paradigms considered in this study are formulated as classification problems, standard classification metrics (accuracy, precision, recall, and F1 score) are used to quantify the extent to which task-related information is preserved after denoising.

b) *Signal Quality Evaluation*: To directly compare the quality of the denoised signals with the reference signals, two metrics are adopted: mean squared error (MSE) and SNR. MSE characterizes the average squared deviation between the denoised and reference signals:

$$\text{MSE}(\hat{\mathbf{X}}, \mathbf{X}_{\text{ref}}) = \frac{1}{CT} \sum_{i=1}^C \sum_{j=1}^T (\hat{\mathbf{X}}(i, j) - \mathbf{X}_{\text{ref}}(i, j))^2 \quad (16)$$

where $\mathbf{X}_{\text{ref}}$ denotes the reference signal, $\hat{\mathbf{X}}(i, j)/\mathbf{X}_{\text{ref}}(i, j)$ is the i th electrode and the j th time point for the denoised signal/reference signal.

The SNR characterizes the relative signal quality by measuring the log-ratio between the root-mean-square amplitude of the reference signal and that of the residual error:

$$\text{SNR}(\hat{\mathbf{X}}, \mathbf{X}_{\text{ref}}) = 20 \lg \left(\frac{\text{RMS}(\mathbf{X}_{\text{ref}})}{\text{RMS}(\hat{\mathbf{X}} - \mathbf{X}_{\text{ref}})} \right) \quad (17)$$

with

$$\text{RMS}(\mathbf{X}) = \sqrt{\frac{1}{CT} \sum_{i=1}^C \sum_{j=1}^T \mathbf{X}(i, j)^2} \quad (18)$$

In all experiments, a lower MSE and a higher SNR indicate stronger denoising performance under the signal-quality metrics. These two metrics are chosen not only because they are commonly adopted in previous EEG denoising studies [14], [32], [40], but also as they capture the denoising effect from two complementary perspectives. Since clean EEG signals are unavailable, following prior work [14], raw EEG recordings are employed as the reference.

3) *Experimental Settings*: To establish a fair comparison, all experiments follow a per-subject evaluation protocol with five-fold cross-validation under the same data split. For each subject, models are trained and evaluated independently; within each fold, the data are partitioned into training, validation, and test sets in a 6:2:2 ratio. For the compared methods, hyperparameter settings follow the recommendation in the original studies. The same data partition and the training epochs are used across all methods to ensure a fair comparison.

For the proposed framework, the batch size is set to 32. For SSVEP, the model is pre-trained for 500 epochs, followed by collaborative optimization for an additional 500 epochs.

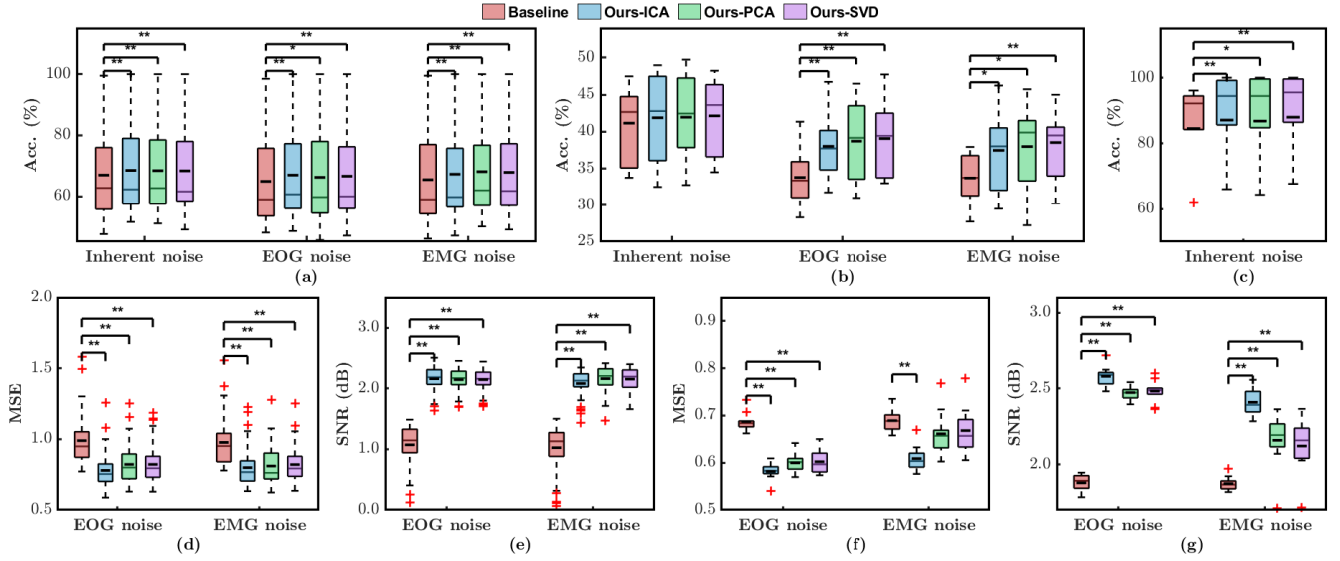


Fig. 2. Comparison of EEG signals before denoising (Baseline) and after applying the proposed task-oriented learning framework with different decompositions (Ours-ICA/PCA/SVD). Task-related information: motor imagery [34] (a), motor execution [36] (b), SSVEP [35] (c). Signal-quality metrics: motor imagery: MSE (d) and SNR (e); motor execution: MSE (f) and SNR (g). Paired Wilcoxon signed-rank tests compare pre- vs. post-denoising; * $p < 0.05$, ** $p < 0.01$.

TABLE I
STRUCTURE OF THE PROPOSED METHOD BASED ON THE EEGNET.

Module	Layer	Kernel	Filters	Padding
	Conv2D	(1, 64)	8	same
	DepthwiseConv.	(Channel, 1)	8×2	—
Feature	Avg. Pooling	(1, 4)	—	—
Extractor	SeparableConv.	(1, 16)	16	same
	Avg. Pooling	(1, 8)	—	—
	Flatten	—	—	—
Classifier	Fully Connected	—	Class	—
Selector	Fully Connected	—	1	—

¹ “Channel”: the number of EEG channels of the input signal.

² “Class”: the number of EEG classes in downstream tasks.

For motor imagery/execution, pre-training and collaborative optimization are each performed for 50 epochs, as further increasing the number of epochs does not yield improvement. During collaborative optimization, AdamW with the AMS-Grad variant is adopted. The learning rate for both the selector and the proxy-task model is initialized at 0.001 and follows cosine decay ($\alpha = 0.9$). Gradients are clipped with a threshold of 1.0. The feature extraction backbone is EEGNet [41]; detailed parameter settings for model structure are reported in Table I, unless otherwise specified.

C. Baseline Methods

1) *Traditional Signal-processing-based Methods*: Traditional methods involve a two-stage process: BSS-based EEG signal decomposition, and informative component selection. Typically, the component selection process is manual. However, to establish a fair comparison with learning-based meth-

ods, representative automatic selection methods for different BSS algorithms are selected in experiments.

- ICA [11]: A network-based classifier for independent component classification.
- PCA [12]: Proposes a filter that delays and orthogonalizes the reference signals to derive a basis, projects the brain sensors onto the noise basis, and removes the projections to yield clean data.
- SVD [22]: Proposes a posteriori estimator, minimizing a global penalty term, to integrate locally denoised segments into globally denoised signals.

2) *Deep-learning-based Methods*: Deep-learning-based methods can be divided into two categories: single channel methods (denoise each EEG channel separately), and multi-channel methods (denoise all EEG channels simultaneously).

a) Single Channel Deep-learning-based Methods:

- 1D-ResCNN [27]: Proposes an end-to-end EEG denoising model based on a one-dimensional residual CNN.
- DuoCL [18]: Proposes a dual-scale CNN-LSTM EEG denoising model for learning spatial-temporal features.
- EEGDNet [17]: Proposes a 2D transformer-based EEG denoising model that considers both non-local and local information in EEG signals.

b) Multi-Channel Deep-learning-based Methods:

- EEGANet [20]: Proposes a GAN-based model for denoising EOG noise in EEG signals.
- STFNet [32]: Proposes an EEG denoising model based on self-attention and stacked multi-dimensional features for learning spatial-temporal representations.
- IC-U-Net [40]: Proposes a U-Net-based EEG denoising model trained by independent components.

D. Experimental Results

1) *Denoising Performance Across Paradigms*: Experiments are conducted on three datasets covering distinct tasks, which

TABLE II

EXPERIMENTAL RESULTS FOR DENOISING INHERENT INSTRUMENTAL AND ENVIRONMENTAL NOISE ON THE MOTOR IMAGERY DATASET [34], THE MOTOR EXECUTION DATASET [36], AND THE SSVEP DATASET [35] (%).

Methods	Motor imagery dataset				Motor execution dataset				SSVEP dataset			
	Acc.	Prec.	Recall	F1	Acc.	Prec.	Recall	F1	Acc.	Prec.	Recall	F1
Baseline	66.99	67.49	67.20	66.45	41.17	40.44	40.98	39.56	84.56	85.06	84.95	82.22
ICA [11]	65.53	66.04	65.73	64.97	38.19	38.25	38.03	36.63	81.54	81.65	81.31	79.52
PCA [12]	67.54	67.97	67.60	67.00	37.14	37.34	37.23	36.02	78.58	79.89	79.46	77.23
SVD [22]	66.39	67.08	66.59	65.61	39.67	39.67	39.67	38.29	84.50	84.87	85.33	83.36
Ours-ICA	68.55	69.03	68.62	68.19	41.90	41.66	41.73	40.40	87.16	88.40	87.41	86.32
Ours-PCA	68.45	68.92	68.56	68.14	41.96	41.54	41.83	40.48	86.85	87.87	87.44	86.03
Ours-SVD	68.39	68.80	68.48	68.09	42.14	41.77	42.03	40.83	88.02	88.78	88.15	87.17

¹ Baseline denotes the raw EEG signal, which is inherently contaminated with instrumental and environmental noise.

TABLE III

EXPERIMENTAL RESULTS FOR DENOISING EOG NOISE ON THE MOTOR IMAGERY DATASET [34] AND THE MOTOR EXECUTION DATASET [36] (%).

Dataset	Model Type	Methods	Task-related Information				Signal Quality	
			Acc. $\uparrow$	Prec. $\uparrow$	Recall $\uparrow$	F1 $\uparrow$	MSE $\downarrow$	SNR $\uparrow$
Motor imagery	Baseline	EEGNet [41]	64.94	65.33	65.05	64.42	0.98	1.07
		ICA [11]	63.17	63.51	63.35	62.56	1.07	0.82
	Traditional	PCA [12]	65.22	65.69	65.36	64.65	1.69	1.50
		SVD [22]	64.44	64.84	64.66	63.49	0.78	2.16
	Single-Channel (DL)	1D-ResCNN [27]	62.92	63.51	63.17	62.15	0.61	3.20
		DuoCL [18]	63.41	63.90	63.60	62.64	0.62	3.24
		EEGDNet [17]	63.21	63.99	63.50	62.20	0.58	3.72
	Multi-Channel (DL)	EEGANet [20]	63.48	63.96	63.66	62.36	0.83	3.22
		STFNet [32]	62.31	62.99	62.56	61.13	0.54	3.08
		IC-U-Net [40]	59.70	59.26	60.21	55.51	1.06	0.73
	Ours	Ours-ICA	66.99	67.44	67.14	66.60	0.77	2.16
		Ours-PCA	66.26	66.64	66.41	65.89	0.82	2.14
		Ours-SVD	66.63	67.07	66.77	66.26	0.82	2.14
Motor execution	Baseline	EEGNet [41]	33.79	34.11	33.77	32.51	0.69	1.88
		ICA [11]	34.31	34.13	34.19	32.66	0.75	0.75
	Traditional	PCA [12]	33.97	34.16	34.09	32.80	1.23	-0.80
		SVD [22]	35.66	35.64	35.63	34.32	0.60	2.39
	Single-Channel (DL)	1D-ResCNN [27]	31.58	30.88	30.91	30.06	0.51	3.10
		DuoCL [18]	36.02	35.75	35.30	34.29	0.67	1.74
		EEGDNet [17]	34.48	33.40	33.69	32.72	0.42	4.09
	Multi-Channel (DL)	EEGANet [20]	36.85	35.91	35.87	34.86	0.43	4.20
		STFNet [32]	38.75	37.64	37.86	37.03	0.43	3.82
		IC-U-Net [40]	37.73	37.37	37.01	35.87	0.64	1.95
	Ours	Ours-ICA	38.02	37.28	37.81	36.65	0.58	2.58
		Ours-PCA	38.72	38.90	38.71	37.73	0.60	2.47
		Ours-SVD	39.08	38.93	39.28	38.03	0.60	2.48

¹ $\uparrow$: indicates that higher values are preferred; $\downarrow$: indicates that lower values are preferred.

² Baseline denotes the raw EEG signal added with EOG noise.

³ DL denotes deep-learning-based algorithms.

are grouped into two paradigms: SSVEP and motor imagery/execution. The task-related information differs between

these paradigms: SSVEP primarily relies on frequency features, whereas motor imagery/execution depends on the com-

TABLE IV

EXPERIMENTAL RESULTS FOR DENOISING EMG NOISE ON THE MOTOR IMAGERY DATASET [34] AND THE MOTOR EXECUTION DATASET [36] (%).

Dataset	Model Type	Methods	Task-related Information				Signal Quality	
			Acc. $\uparrow$	Prec. $\uparrow$	Recall $\uparrow$	F1 $\uparrow$	MSE $\downarrow$	SNR $\uparrow$
Motor imagery	Baseline	EEGNet [41]	65.47	66.02	65.62	64.88	0.97	1.02
		ICA [11]	65.33	65.80	65.46	64.70	1.02	0.86
		PCA [12]	66.65	67.00	66.76	66.10	1.36	0.30
		SVD [22]	65.81	66.49	66.01	64.91	0.78	2.12
	Single-Channel (DL)	1D-ResCNN [27]	63.92	64.43	64.15	63.14	0.56	3.42
		DuoCL [18]	64.70	65.43	64.86	64.04	0.49	4.27
		EEGDNet [17]	65.16	65.92	65.35	64.47	0.46	4.72
	Multi-Channel (DL)	EEGANet [20]	64.64	65.30	64.76	63.48	0.77	4.02
		STFNet [32]	63.38	63.75	63.55	62.43	0.38	3.53
		IC-U-Net [40]	60.01	59.99	60.39	56.49	1.04	0.93
	Ours	Ours-ICA	67.31	67.84	67.43	66.92	0.80	2.08
		Ours-PCA	68.11	68.69	68.26	67.72	0.81	2.16
		Ours-SVD	67.87	68.45	67.99	67.46	0.82	2.15
Motor execution	Baseline	EEGNet [41]	33.72	32.82	33.63	32.09	0.69	1.87
		ICA [11]	36.06	36.03	35.94	34.74	0.75	1.59
		PCA [12]	34.29	34.00	34.35	32.97	1.23	-0.80
		SVD [22]	35.87	34.83	35.57	34.05	0.60	2.41
	Single-Channel (DL)	1D-ResCNN [27]	31.70	31.58	31.65	30.01	0.52	3.13
		DuoCL [18]	35.99	35.55	35.73	34.46	0.68	1.82
		EEGDNet [17]	34.79	34.59	34.70	33.09	0.44	4.07
	Multi-Channel (DL)	EEGANet [20]	36.81	36.66	36.74	35.33	0.44	4.18
		STFNet [32]	37.19	36.55	37.00	35.28	0.44	3.82
		IC-U-Net [40]	36.00	35.26	35.91	33.94	0.65	1.99
	Ours	Ours-ICA	37.48	37.70	37.37	36.19	0.61	2.41
		Ours-PCA	37.98	38.02	38.07	36.69	0.66	2.16
		Ours-SVD	38.53	38.51	38.49	37.11	0.67	2.12

¹ $\uparrow$: indicates that higher values are preferred; $\downarrow$: indicates that lower values are preferred.

² Baseline denotes the raw EEG signal added with EMG noise.

³ DL denotes deep-learning-based algorithms.

prehensive frequency-spatial-temporal representations. This setting tests whether the task-oriented learning framework accommodates various signal characteristics.

Results (Tables II, III, and IV) indicate that the task-oriented learning framework works across paradigms (SSVEP, motor imagery, and motor execution). Because ocular and muscular artifacts typically have limited impact on signals recorded over the occipital cortex in SSVEP, only inherent noise is considered for that paradigm. Improvement in task-related information can be observed (Acc.: 2.78% $\uparrow$, Prec.: 3.29% $\uparrow$, Recall: 2.72% $\uparrow$, F1: 4.29% $\uparrow$). For motor imagery/execution, average gains across inherent, EOG, and EMG noise conditions are as follows (motor imagery/motor execution): Acc.: 1.82%/3.31% $\uparrow$, Prec.: 1.82%/3.56% $\uparrow$, Recall: 1.78%/3.35% $\uparrow$, F1: 2.00%/3.51% $\uparrow$. Signal-quality metrics under physiological noise conditions (motor imagery/execution) also improve: MSE decreases by 0.17/0.07, while SNR increases by 1.10/0.50 dB.

These results indicate that the proposed task-oriented denoising framework, trained using downstream task labels, can consistently identify informative components that support task performance and improve signal quality across diverse EEG characteristics. This observation suggests that the framework may serve as a general-purpose denoising approach applicable to multiple EEG paradigms.

2) *Denoising Performance Across Noise*: EEG signals can be contaminated by different noise sources. The proposed task-oriented framework is evaluated on both simulated physiological artifacts (EOG, EMG) and inherent instrumental/environmental noise, and consistently shows improvements in signal quality under these conditions.

For inherent noise, average task-related metrics improve from the baseline over three datasets can be found: Acc.: 1.69% $\uparrow$, Prec.: 1.96% $\uparrow$, Recall: 1.65% $\uparrow$, F1: 2.42% $\uparrow$. For simulated artifacts, evaluated on the motor imagery and execution datasets, gains are observed in both task-related met-

rics and signal-quality measures. Task-related metrics improve as follows: Acc.: 3.26%/3.29% $\uparrow$, Prec.: 2.99%/3.79% $\uparrow$, Recall: 3.28%/3.31% $\uparrow$, F1: 3.40%/3.50% $\uparrow$. Signal-quality metrics also improve: MSE decreases by 0.14/0.10, while SNR increases by 0.86/0.74 dB.

Figure 2 summarizes subject-level distributions of evaluation metrics for baseline and denoised signals across datasets and noise conditions. On average, all metrics improve after denoising. Moreover, most comparisons show statistically significant gains according to paired Wilcoxon signed-rank tests ($p < 0.05$; $p < 0.01$), with only 5 non-significant cases out of 45 tests. The analysis under Wilcoxon signed-rank tests shows that the proposed task-oriented learning framework consistently improves signal quality across experimental paradigms and noise types. Such generalization may be attributed to the task-oriented design, in which task labels guide denoising and encourage emphasis on structured task-related information rather than irregular noise components.

3) *Competitive Performance for EEG Denoising*: To comprehensively assess performance, the proposed task-oriented learning framework is compared with representative EEG denoising methods spanning traditional signal-processing-based [11], [12], [22] and learning-based methods [17], [18], [20], [27], [32], [40]. A brief description of these baselines is provided in Sec. IV-C. Traditional BSS-based approaches include ICA, PCA, and SVD; these decomposition techniques are also adopted within the proposed framework. Learning-based approaches cover a range of architectures (e.g., CNNs, RNNs, and transformers) and design settings (e.g., single- and multi-channel denoising).

Deep-learning-based methods achieve lower MSE and higher SNR on signal-quality metrics, as reported in Tables III and IV. This behavior is expected, since their training objectives are explicitly aligned with these criteria (e.g., minimizing MSE or matching the distribution of clean reference signals). However, task-relevant information may be degraded during training, as indicated by reduced performance on proxy downstream classification tasks. For example, on the motor imagery dataset with EOG/EMG artifacts [34], signals denoised by learning-based baselines yield lower downstream performance than the original noisy inputs. These observations indicate that gains in MSE/SNR may not directly translate into improvements for applications.

Moreover, because truly clean EEG references are unavailable for inherent instrumental and environmental noise, supervised learning methods that require clean targets have been excluded from the comparison in Table II. This underscores a key advantage of the proposed task-oriented learning framework: training relies solely on readily available proxy-task labels, enabling applicability across diverse noise sources, including those inherent to EEG.

Traditional signal-processing-based methods often underperform on signal quality metrics yet cause limited degradation in task-related accuracy. This reflects a strength (minimal artifact introduction through component selection) but also a weakness (the difficulty of reliably selecting informative components). The proposed task-oriented learning framework leverages the robustness of decomposition while mitigating the

TABLE V
COMPARISON OF MODELS TRAINED WITH THE LEARNED SELECTOR VERSUS UNIFORM RANDOM COMPONENT MIXING ON THE SSVEP DATASET [35] (%).

Algorithm	Mode	Acc.	Prec.	Recall	F1
Baseline		84.56	85.06	84.95	82.22
ICA	Random	81.29	81.68	81.47	79.20
	Selector	87.16	88.40	87.41	86.32
PCA	Random	82.40	82.48	83.33	80.59
	Selector	86.85	87.87	87.44	86.03
SVD	Random	83.58	83.57	84.00	81.95
	Selector	88.02	86.78	88.15	87.17

¹ Baseline denotes the raw EEG signal, which is inherently contaminated with instrumental and environmental noise.

challenge of selection via a selector trained under task-related label supervision, delivering improvements in signal quality and preservation of task-relevant information.

In summary, the proposed task-oriented learning framework demonstrates clear advantages over the compared methods. Although deep-learning-based approaches often achieve lower MSE and higher SNR on signal-quality metrics, they require clean signals for training, and the supervision process may attenuate task-related information in EEG signals. In contrast, the proposed framework does not require clean reference signals. Guided by task labels, it explicitly learns task-relevant information from the decomposed components, thereby improving task-related performance after denoising. Compared with conventional signal-processing-based methods, the proposed framework employs a learning-based selector supervised by task labels to replace manual component inspection, improving signal quality.

4) *Retention of Task-related Information during Denoising*: The motivation for EEG denoising is to facilitate analysis of task-related information. Hence, a critical criterion is whether task-relevant content is preserved after denoising. Because task-related components are often weak and entangled with noise, signal-quality metrics may not fully capture their integrity. A proxy-task classifier is employed to quantify task-relevant information in the denoised EEG. As summarized in Tables II, III, and IV, the proposed task-oriented learning framework exhibits consistent advantages. Considering accuracy (averaged over the compared methods), the gains are:

- **SSVEP**: 2.78%/2.84% $\uparrow$ (inherent).
- **Motor imagery**: 1.47%/0.92% $\uparrow$ (inherent), 1.69%/1.41% $\uparrow$ (EOG), 2.29%/1.11% $\uparrow$ (EMG).
- **Motor execution**: 0.83%/2.33% $\uparrow$ (inherent), 4.82%/0.14% $\uparrow$ (EOG), 4.28%/0.81% $\uparrow$ (EMG).

Here, each pair denotes improvement over the baseline / over the best competing method. Note that the strongest compared method differs across settings; nevertheless, the proposed pipeline delivers the most consistently superior accuracy, indicating that the denoising process preserves and makes more salient task-related information.

TABLE VI
COMPARISON OF COLLABORATIVE OPTIMIZATION WITH STABILIZATION TECHNIQUES (OURS) VERSUS THE BASELINE WITHOUT (BL) (%).

Noise	Dataset	Mode	Acc.	Prec.	Recall	F1
Inherent	MI	BL	68.19	68.58	68.26	67.86
		Ours	68.46	68.92	68.55	68.14
	ME	BL	42.24	42.01	42.24	40.95
		Ours	42.00	41.66	41.86	40.57
	SSVEP	BL	86.11	86.15	86.51	84.56
		Ours	87.34	88.35	87.67	86.51
EOG	MI	BL	66.71	67.10	66.82	66.36
		Ours	66.63	67.05	66.77	66.25
	ME	BL	38.92	38.44	38.84	37.57
		Ours	38.61	38.37	38.60	37.47
EMG	MI	BL	67.58	67.97	67.68	67.24
		Ours	67.76	68.33	67.89	67.37
	ME	BL	36.53	36.36	36.59	35.43
		Ours	38.00	38.08	37.98	36.66

¹ Results in this table are averaged across decomposition methods (PCA, ICA, SVD).

² MI: motor imagery; ME: motor execution.

E. Ablation Study

1) *The Component Selection Strategy of Selector*: The change of the selector's probabilistic outputs during the training introduces diversity in reconstructed EEG signals, which could potentially act as a data augmentation module. This raises the question of whether observed improvements in decoder performance stem from increased data diversity or from enhanced signal quality due to denoising. To disentangle these effects, an ablation study is conducted on the SSVEP dataset [35]: comparison decoders are trained on EEG reconstructed by combining decomposed components with coefficients sampled uniformly at random.

As shown in Table V, relative to the baseline trained on raw EEG, the random-mixing augmentation degrades performance (Acc.: 2.14% ↓, Prec.: 2.48% ↓, Recall: 2.02% ↓, F1: 1.64% ↓). In contrast, reconstruction guided by the learned selector improves performance (Acc.: 2.78% ↑, Prec.: 3.29% ↑, Recall: 2.72% ↑, F1: 4.29% ↑). These findings indicate that naively varying component ratios does not constitute an effective data-augmentation strategy; the gains arise from improved EEG quality rather than diversity alone.

2) *Stabilization Techniques for Collaborative Optimization*: To help satisfy the hypotheses (H1-H4) in Sec. III-C2, several optimization techniques are employed, including gradient clipping and AdamW with the AMSGrad variant. Although these techniques do not guarantee that the hypotheses hold, they encourage the optimization dynamics to conform to the stated conditions. To assess their contribution, an ablation compares training with and without these techniques. For robustness, the results in Table VI are averaged across three decomposition methods (ICA, PCA, SVD).

TABLE VII
ABLATION OF THE SHARED FEATURE EXTRACTOR FOR THE CLASSIFIER AND SELECTOR ON THE MOTOR EXECUTION DATASET [36] (%)

Method	Acc.	Prec.	Recall	F1	MSE	SNR
Baseline	33.79	34.11	33.77	32.51	0.69	1.88
ICA + NS	39.01	38.69	38.66	37.72	0.65	2.17
ICA + S	38.02	37.28	37.81	36.65	0.58	2.58
PCA + NS	37.45	37.75	37.38	36.27	0.71	1.76
PCA + S	38.72	38.90	38.71	37.73	0.60	2.47
SVD + NS	36.53	37.25	36.22	35.10	0.71	1.68
SVD + S	39.08	38.93	39.28	38.03	0.60	2.48

¹ Baseline denotes EEGNet trained on raw EEG signals contaminated with EOG noise. "NS" denotes the feature extractor is *not shared* between the classifier and selector, whereas "S" denotes the feature extractor is *shared*.

TABLE VIII
SENSITIVITY ANALYSIS OF THE NUMBER OF DECOMPOSITION COMPONENTS ON THE MOTOR EXECUTION DATASET [36] (%)

Components	Acc.	Prec.	Recall	F1	MSE	SNR
2	38.99	38.37	39.03	37.60	0.60	2.51
4	39.29	38.88	39.03	37.93	0.61	2.42
8	37.66	37.28	37.53	36.39	0.60	2.49
16	38.92	38.58	38.90	37.45	0.58	2.57
32 (Ours)	38.02	37.28	37.81	36.65	0.58	2.58

Across four of seven evaluation conditions, the stabilization techniques improve performance relative to training without them. Furthermore, the gains (Acc.: 0.79% ↑; Prec.: 1.16% ↑; Recall: 0.76% ↑; F1: 0.90% ↑) exceed the occasional declines in magnitude (Acc.: 0.21% ↓; Prec.: 0.16% ↓; Recall: 0.22% ↓; F1: 0.20% ↓). These findings indicate marginal but consistent benefits for collaborative training. The modest overall effect likely reflects the task-oriented learning framework's intrinsic adherence to the stated assumptions, which limit the potential for further improvement from the optimization.

3) *Shared Feature Extractor for Classifier and Selector*: The use of a shared feature extractor for the classifier and selector is motivated by the task-oriented design, in which denoising is formulated as identifying task-relevant information from the decomposed components. To validate this design choice, an ablation study is conducted on the motor execution dataset corrupted with EOG noise, comparing shared and non-shared extractors under different BSS decompositions (ICA/PCA/SVD). As summarized in Table VII, introducing the shared extractor improves signal-quality metrics across the BSS methods used in the decomposition stage. For task-related metrics, the use of the shared extractor increases performance for all decomposition methods except ICA. Moreover, the shared extractor configuration reduces the computational cost during training. These results suggest that a shared representation is sufficient for both task classification and noise identification, supporting the rationale of the proposed design.

4) *Sensitivity Analysis of the Number of Decomposition Components*: The number of components used in the decomposition is an important parameter of the framework. In the

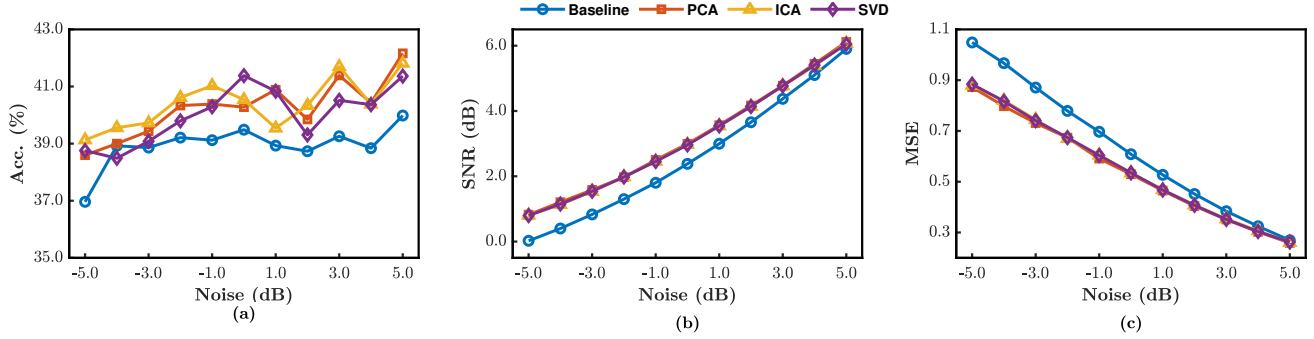


Fig. 3. Performances on the motor-execution dataset [36] across noise levels.

TABLE IX
EXPERIMENTAL RESULTS FOR THE PROPOSED TASK-ORIENTED LEARNING FRAMEWORK COMBINED WITH DIFFERENT LEARNING-BASED FEATURE EXTRACTORS ON THE SSVEP DATASET [35] (%).

Methods	EEGNeX [42]				EEGTCNet [43]				DeepConvNet [44]			
	Acc.	Prec.	Recall	F1	Acc.	Prec.	Recall	F1	Acc.	Prec.	Recall	F1
Baseline	89.93	90.45	90.13	89.13	70.37	70.93	70.84	68.95	81.11	81.72	80.97	79.10
ICA [11]	85.80	85.52	85.62	84.05	64.75	64.72	64.84	62.76	71.35	72.29	71.89	69.64
PCA [12]	85.74	86.56	86.43	85.30	62.40	63.23	62.30	60.08	65.80	67.04	65.47	63.33
SVD [22]	85.24	85.24	86.56	86.15	69.38	69.35	69.87	67.36	78.14	80.47	78.55	77.05
Ours-ICA	92.90	92.82	92.70	92.20	77.34	76.87	77.18	75.42	92.71	93.35	92.95	92.20
Ours-PCA	92.59	92.80	92.40	91.99	76.60	76.16	76.39	74.68	92.83	94.04	93.23	92.63
Ours-SVD	92.65	92.74	92.55	92.04	76.72	76.59	76.52	74.83	93.14	93.66	93.34	92.59

¹ Baseline denotes the raw EEG signal, which is inherently contaminated with instrumental and environmental noise.

current setting, this number is chosen to match the number of EEG channels, consistent with decomposition analysis in EEG. However, since the component count can affect the decomposition outcome and the denoising, sensitivity analysis is necessary. An ablation study is conducted on the motor execution dataset corrupted with EOG noise under ICA decomposition. The dataset is collected by a 32-channel EEG cap, and the number of components is set to [2, 4, 8, 16, 32]. As reported in Table VIII, varying the component count yields a limited impact on the denoising performance. A possible explanation is that signal reconstruction in the proposed framework relies on the soft-threshold weighting, which can adaptively adjust the contributions of different components. Based on these results, setting the number of components equal to the number of channels remains a safe and practical choice, as it aligns with decomposition-based denoising settings in EEG processing.

V. DISCUSSION

A. Performance Under Varying Noise Levels

To assess robustness to different noise levels, an additional experiment is conducted on a motor execution dataset [36] with synthetic EOG artifacts added at SNR ranging from -5 to 5 dB (Fig. 3). Across all SNRs, both signal-quality metrics and task-related metrics of the denoised EEG improve relative to the noisy baseline. The gains in signal-quality metrics are larger at more severe noise (e.g., near -5 dB) than at milder noise (e.g., near 5 dB), whereas improvements in classification

TABLE X
EXPERIMENTAL RESULTS FOR DENOISING COMBINED EOG AND EMG NOISE ON THE MOTOR EXECUTION DATASET [36] (%).

Methods	Acc.	Prec.	Recall	F1	MSE	SNR
Baseline	30.88	30.33	30.73	29.36	1.16	-0.45
Ours-ICA	35.78	35.80	35.52	34.39	0.97	0.33
Ours-PCA	34.97	34.68	34.91	33.74	0.98	0.33
Ours-SVD	35.69	35.63	35.50	34.59	0.99	0.32

¹ Baseline denotes the raw EEG signal with EOG and EMG noise.

accuracy are comparatively uniform across SNRs. This pattern suggests that the task-guided selector can suppress not only the synthetic EOG artifacts but also the inherent noise that is not explicitly simulated, which may not be fully reflected by the signal-quality metrics.

B. Performance Under the Combined EMG and EOG Noise

In practice, EEG recordings may be simultaneously affected by multiple noise sources. To reflect this setting, an experiment is conducted on the motor execution dataset, where each EEG trial is contaminated with both EMG and EOG noise. The SNR values of the injected EMG and EOG artifacts are drawn uniformly from $[-5, 5]$ dB. As reported in Table X, consistent improvements are observed under different BSS algorithms in both task-related metrics (Acc.: 4.6% $\uparrow$, Prec.: 5.04% $\uparrow$,

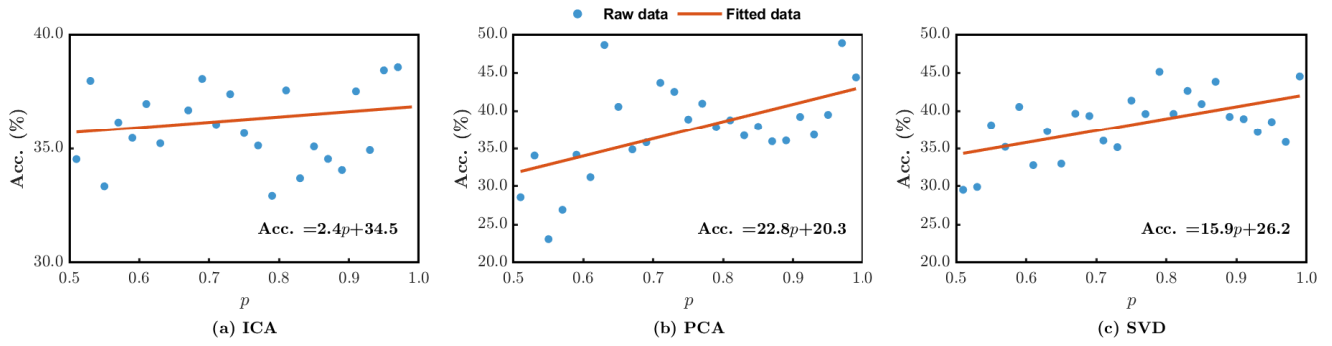


Fig. 4. Component-level classification accuracy on the SSVEP dataset [35] across BSS algorithms.

Recall: 4.58% $\uparrow$, and F1: 4.88% $\uparrow$) and signal-quality metrics (MSE: 0.18 $\downarrow$, SNR: 0.78 $\uparrow$). These results indicate that the proposed framework remains effective when denoising EEG signals corrupted by combined noise.

C. The Framework is Algorithm-agnostic

The proposed framework involves two parts: (1) the BSS-based decomposition module and (2) the selector and proxy-task models. To validate that the proposed framework is algorithm-agnostic, various decomposition algorithms and learning-based models are evaluated within the framework [41]–[44] (Table IX). The decomposition algorithms can be directly applied within the framework, while the learning-based models keep the representation learning module, changing the dimension of the output head for the designed tasks.

For the BSS decomposition module, three widely used algorithms are evaluated across different tasks. Experimental results across multiple decomposition methods demonstrate improvements in both task-related and signal-quality metrics relative to traditional denoising pipelines, indicating that the proposed task-oriented learning framework is compatible with different decomposition algorithms.

For the selector and proxy-task models, an additional study is conducted on the SSVEP dataset [35] using several network architectures [41]–[44]. Although performance varies across architectures, all demonstrate improved downstream classification compared to the original noisy EEG signals, implying that noisy components are suppressed while task-related components are emphasized. This behavior aligns with the motivation for denoising and further confirms that the framework operates in an algorithm-agnostic manner.

D. The Impact of the Selector in the Framework

1) *Selector Favors Task-related Components*: The framework is designed to select task-related components from the decomposition, which are expected to be more identifiable in the proxy tasks. To assess this hypothesis, a component-level classifier is trained. As shown in Fig. 4, classification accuracy increases as the selector's retention probability p (i.e., the probability assigned to keep a component) increases. This trend indicates that components favored by the selector are more informative for the proxy task, supporting the idea that the selector prioritizes task-related components over noise.

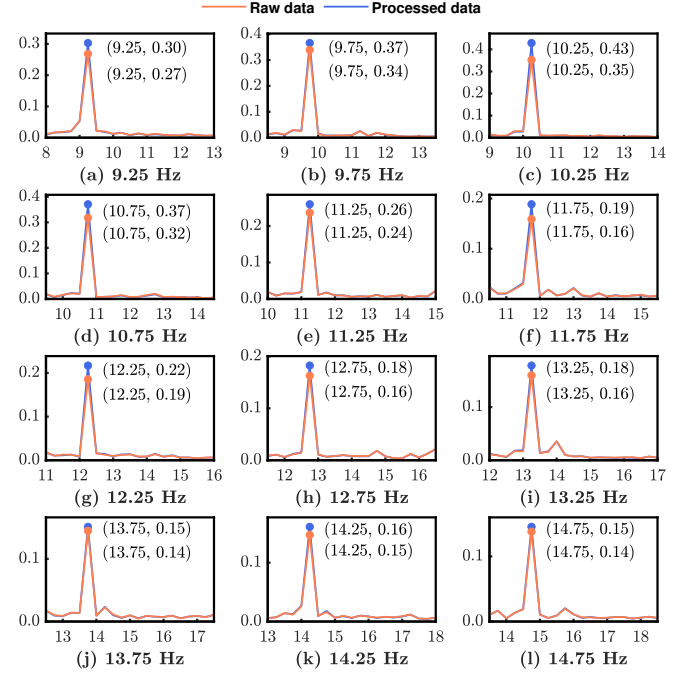


Fig. 5. Comparison of power spectra and per-frequency power ratios for raw EEG and EEG processed by the proposed task-oriented learning framework using ICA-based decomposition for a representative subject from the SSVEP dataset [35]. Panels (a)–(l) correspond to stimulus fundamental frequencies (9.25–14.75 Hz in 0.5 Hz steps); markers denote the peak magnitude at each target frequency.

2) *Processed Data Exhibits Higher Fundamental-Frequency Power in the SSVEP Paradigm*: To examine why the denoised signals are easier to classify, the power spectrum and per-frequency power ratios are visualized. Relative to the raw recordings, the processed EEG exhibits a higher power ratio at the stimulus fundamental frequency (Fig. 5). This enhancement is crucial for the SSVEP paradigm, indicating that denoising preserves task-relevant oscillatory components while suppressing irrelevant activity.

3) *Processed Data Exhibits More Coherent Spectrotemporal Features in the Motor Imagery Paradigm*: In motor imagery, the dominant EEG characteristic is the modulation of the sensorimotor rhythm (SMR), which is observed as ERD/ERS. Baseline variability introduced by EOG contamination complicates the direct comparison of ERD/ERS wave-

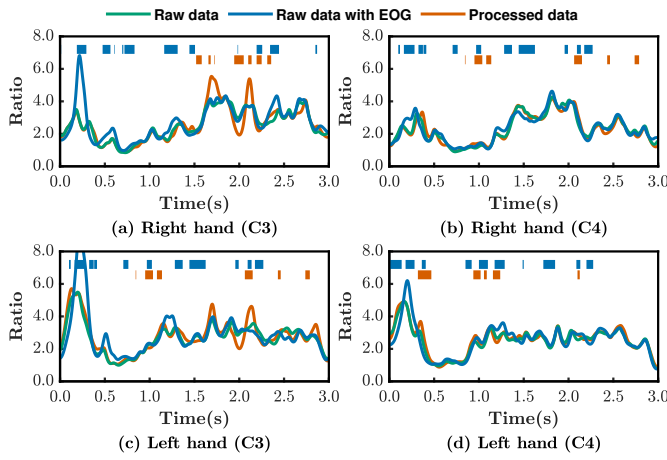


Fig. 6. Alpha-to-beta (8-12 Hz / 12-30 Hz) power-ratio trajectories for raw EEG, raw EEG with added EOG, and EEG processed by the proposed task-oriented learning framework using ICA-based decomposition for a representative subject from the motor imagery dataset [34]. Colored significance bars along the top margin indicate time points significantly different from the raw baseline (paired two-tailed t -test): blue bars (upper track) for raw+EOG vs. raw, and orange bars (lower track) for processed vs. raw ($p < 0.05$).

forms between reference and denoised signals. Therefore, the alpha-to-beta power ratio is employed as a measure of SMR modulation, focusing on the alpha (8-12 Hz) and beta (12-30 Hz) bands most implicated in ERD/ERS.

A case study on the motor imagery dataset [34] (Fig. 6) visualizes the alpha-to-beta power ratio under three conditions: raw data, raw data with added EOG, and the processed (denoised) signal. The figure is averaged over twenty trials for channels C3 and C4, which are closely associated with motor cortex activity. Relative to the raw baseline, EOG contamination distorts the ratio trajectory, whereas the processed EEG recovers a trajectory that closely follows the raw pattern.

A time-point-wise t -test indicates that EOG-contaminated signals exhibit substantially more significant deviations from the raw data than the processed signals. This finding suggests that the proposed denoising procedure restores the underlying power distribution and suppresses artifact-related components, which is crucial for analyzing motor imagery.

E. Limitations and Future Work

Although the proposed framework demonstrates benefits for EEG denoising, several limitations remain. First, while task-related labels are easier to obtain than the clean EEG signals required by many learning-based denoising methods, labels may still be unavailable in certain scenarios (e.g., mental-state monitoring). This limitation motivates extensions to label-free or unsupervised training settings. Second, the current framework relies on conventional decomposition algorithms. Since these methods may not be specifically tailored to EEG characteristics, further performance improvements may be possible through EEG-adaptive decomposition. Another limitation is that the proposed method follows a two-stage pipeline, with a BSS-based decomposition followed by collaborative optimization. This separation increases pipeline complexity. Moreover, it prevents task-related supervision from being

exploited during the decomposition stage, even though such supervision could potentially guide the decomposition toward task-relevant components.

These limitations suggest several directions for future research. One direction is to develop a denoising framework under label-free settings. A potential solution is to train the selector via unsupervised representation learning (e.g., contrastive learning or masked autoencoders), followed by clustering in the learned feature space to separate noise-related components from informative components. Another direction is to incorporate a trainable decomposition module, e.g., differentiable BSS techniques [45], to unify the current two-stage pipeline into an end-to-end framework to reduce pipeline complexity. This design can also streamline training by enabling task-oriented supervision to guide the decomposition stage, encouraging an explicit separation between task-relevant and task-irrelevant components.

VI. CONCLUSIONS

This study presents a task-oriented learning framework for automatic EEG denoising that uses task labels without clean reference EEG signals. The procedure decomposes the EEG into components with BSS methods, applies a learning-based selector to assign retention probabilities, and reconstructs a denoised signal as a probability-weighted combination. The supervision is achieved by a proxy-task model under a collaborative optimization scheme that relies solely on task labels. The compatibility of the task-oriented learning framework with multiple decomposition strategies and network backbones indicates broad applicability. These properties enable practical denoising without manual intervention or clean references and suggest an impact for EEG-based brain-computer interfaces and neuroscience research. Moreover, the task-oriented learning framework may be extended to other bio-signal modalities for denoising.

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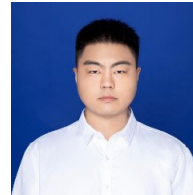
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